

Q4 Seasonality Statistical Analysis

Online Retail II Data Pipeline Project

Overview

After building the analytics layer, I wanted better to understand the seasonal patterns in the revenue data. When I plotted daily revenue over time, I noticed that revenue tends to increase significantly during the fourth quarter (October through December).

To verify whether this seasonal increase was statistically significant, I performed a series of statistical tests using daily-aggregated data from the warehouse's analytics layer.

The goal was to answer two main questions:

1. Is daily revenue during Q4 statistically higher than during the rest of the year?
 2. If revenue is higher during Q4, is the increase caused by higher spending per order or simply by higher transaction volume?
-

Exploratory Analysis

Before running any formal statistical tests, I examined the distribution of daily revenue over time.

Time-series plots showed a clear pattern where revenue spikes repeatedly during the fourth quarter across multiple years in the dataset. To investigate this pattern more closely, I separated the daily observations into two groups:

- Q4 observations (October, November, December)
- Non-Q4 observations (all other months)

I also created boxplots to compare the distributions of daily revenue between these two groups.

The boxplots showed several noticeable differences:

- The median daily revenue during Q4 is higher.
- The interquartile range is wider for Q4 observations.
- The spread of revenue values appears larger during Q4.

These differences suggested that the two groups might have unequal variances. Because of this, I decided to use Welch's two-sample t-test, which does not assume equal variances between groups.

Hypothesis Testing

To formally test whether revenue differs between Q4 and the rest of the year, I performed a Welch two-sample t-test using the daily revenue values.

The hypotheses were defined as follows:

Null hypothesis (H_0):

The mean daily revenue during Q4 is equal to the mean daily revenue during non-Q4 months.

Alternative hypothesis (H1):

The mean daily revenue during Q4 is greater than the mean daily revenue during non-Q4 months.

The test results showed:

- Average Q4 daily revenue is about \$21,660 higher
- p-value < 0.001
- 95% confidence interval:
\$17,953 to \$25,368

Because the p-value is well below the 0.05 significance level, there is sufficient statistical evidence to reject the null hypothesis. The results indicate that average daily revenue during Q4 is higher than during the rest of the year.

Return Rate Analysis

I also tested whether the return rate behaves differently during Q4.

The same Welch testing framework was used to compare return rates between Q4 and non-Q4 periods.

The results showed that the difference in return rates was not statistically significant ($p = 0.52$).

This suggests that the increase in Q4 revenue is not caused by abnormal return behavior.

In other words, the seasonal revenue spike appears to reflect genuine increases in sales activity rather than unusual return patterns.

Initial Regression Model

To further quantify the seasonal effect, I estimated a simple regression model using a binary indicator variable for Q4.

The model can be written as:

$$\text{Daily Revenue} = \beta_0 + \beta_1(\text{Q4 Indicator})$$

In this model:

- The dependent variable is daily revenue.
- The independent variable is a Q4 indicator, which equals 1 if the observation occurs in Q4 and 0 otherwise.

The estimated coefficient for the Q4 indicator was approximately:

\$21,660

This coefficient was highly statistically significant. The model's R-squared value was about 0.266, meaning that the Q4 indicator alone explains roughly 26.6% of the variation in daily revenue.

While this confirms the seasonal pattern, this model does not account for differences in transaction volume.

Controlled Regression Model

To better understand the underlying cause of the seasonal revenue increase, I estimated a second regression model that includes daily order count as a control variable.

The model can be written as:

$$\text{Daily Revenue} = \beta_0 + \beta_1(\text{Q4 Indicator}) + \beta_2(\text{Daily Orders})$$

In this model:

- The dependent variable remains daily revenue.
- The independent variables include both the Q4 indicator and the number of daily orders.

The results showed:

- Daily Orders coefficient: about \$15.37 per additional transaction
 - Q4 coefficient: reduced to about \$1,755
 - The Q4 coefficient was no longer statistically significant ($p = 0.21$).
 - R-squared increased to about 0.600, meaning the model explains around 60% of the variation in daily revenue.
-

Interpretation of the Results

Comparing the two regression models provides an important insight. In the first model, the Q4 indicator appears to predict higher revenue strongly. However, once the daily order count is included in the model, the estimated effect of Q4 becomes much smaller and is no longer statistically significant. This suggests that the seasonal increase in revenue during Q4 is largely explained by higher transaction volume rather than higher spending per order. In other words, customers are not necessarily spending more per purchase during Q4. Instead, the higher revenue levels observed during this period appear to be driven by an increase in transaction volume.

Final Insight

The statistical analysis confirms that the Q4 revenue spike observed in the data is real. However, the increase is primarily driven by higher customer activity rather than structural changes in spending behavior.

This distinction is important because it changes how the seasonal pattern should be interpreted. The data suggests that the business experiences a surge in transaction volume during the holiday season, resulting in higher revenue.

Conclusion

This analysis demonstrates how the analytics layer built from the data pipeline can support deeper statistical investigation.

By combining exploratory analysis, hypothesis testing, and regression modeling, I confirmed that the Q4 revenue increase is statistically significant and explained the mechanism behind it.

The results show that the seasonal spike is mainly driven by increased transaction activity rather than higher spending per order. This type of insight would be useful for planning inventory levels, forecasting demand, and understanding seasonal customer behavior.